

AI-Driven Innovations: Edge-to-Cloud Intelligence

COMPUTEX 2025 was a showcase for the accelerated integration of artificial intelligence into both consumer and industrial ecosystems. One of the most compelling narratives was the seamless collaboration between edge and cloud computing, a trend that promises to unlock new possibilities across every sector.



MediaTek's 5G Generative AI Gateway stood out as a prime example. This innovative platform allows devices at the edge—such as smartphones, smart home devices, and industrial sensors—to perform complex AI tasks locally while still leveraging the vast computational power of the cloud for more demanding workloads. MediaTek's demonstration highlighted how real-time language translation, advanced image recognition, and predictive maintenance can be handled efficiently, reducing latency and improving privacy by keeping sensitive data on local devices.

NVIDIA's Collaboration with Edge and Cloud Partners was another highlight. The company showcased how its AI technologies are being integrated into manufacturing processes, ena-

bling factories to become more adaptive and intelligent. By combining edge AI with cloud analytics, manufacturers can now predict equipment failures, optimize production lines, and reduce downtime—all in real time.

AI in Industry: Beyond the Factory Floor

The industrial applications of AI were on full display, with companies like Advantech and Pegatron leading the charge. Advantech's Autonomous Mobile Robots (AMRs) were a crowd favourite, demonstrating how AI-powered robots can navigate complex environments, transport materials, and even collaborate with human workers. These AMRs are equipped with advanced sensors and machine learning algorithms, allowing them to adapt to dynamic factory layouts and avoid obstacles with ease.

Pegatron's biomimetic robot dogs were another standout, showcasing the potential for AI-driven robotics in both industrial and consumer settings. These robotic companions can perform tasks ranging from surveillance and inspection to providing assistance in healthcare and hospitality. Their lifelike movements and ability to learn from their environment drew crowds and sparked conversations about the future of human-robot interaction.



BenQ's AI Golf Simulator demonstrated how AI is making its way into sports and entertainment. The simulator uses computer vision and machine learning to analyse a player's swing, providing real-time feedback and personalized coaching tips. This application of AI not only enhances the user experience but also opens up new possibilities for training and performance analysis in other sports. **d**

